ROHIT SHENDURKAR

rohitshendurkar2017@gmail.com | +91-8421252700 | LinkedIn | GitHub

EDUCATION

Indian Institute of Information Technology, Nagpur **2022 – 2026**

Bachelor of Technology in Electronics and Communication Engineering *Nagpur, India*

EXPERIENCE

Full-Stack Developer Intern **Aug 2025 – Aug 2026**

Elixir Strategic Management *Nagpur, India*

- Engineered backend services for an industrial IoT platform that ingests real-time sensor telemetry, computes operational KPIs, and serves analytics through RESTful APIs to multi-tenant dashboards — built with Python, Node.js, Express.js, and MongoDB.
- Designed authentication, device management, and automated alert APIs, establishing the core service layer that connects sensor data pipelines to production-grade monitoring dashboards.
- Restructured MongoDB aggregation pipelines for time-series telemetry workloads, reducing query latency and enabling near-real-time analytics across thousands of concurrent industrial data points.

Software Development Intern **Mar 2025 – Jul 2025**

Modgenics Solutions India

- Architected RESTful API endpoints and integrated token-based authentication using Node.js and MongoDB, enforcing role-based access control across a multi-user internal dashboard application.
- Developed data-driven dashboard views in React.js backed by paginated API queries, replacing manual spreadsheet workflows and cutting data retrieval time for operations teams.

PROJECTS

PowerIQ-AI | Industrial IoT Energy Intelligence Platform **2025**

- Designed a modular FastAPI backend to ingest industrial sensor telemetry, compute real-time KPIs for energy consumption, cost forecasting, and power quality metrics, and expose results through versioned REST endpoints serving interactive dashboards.
- Implemented threshold-based alerting and device health monitoring APIs that track electrical parameters, solar utilization, and demand patterns — enabling operators to detect anomalies before they escalate to unplanned downtime.

Real-Time Traffic Speed Monitoring System **2024**

- Built a computer vision pipeline combining YOLOv8 object detection with SORT multi-object tracking and dual-line speed estimation geometry, calculating vehicle speeds from frame-level timestamp deltas with sub-second inference latency.
- Automated violation detection and persistence — the system logs overspeeding events with vehicle ID, computed speed, and annotated frame snapshots, processing live video feeds end-to-end using Python, OpenCV, and TensorFlow.

RAG Pipeline | Gemini + Qdrant Vector DB + Cohere Reranker **2025**

- Architected a retrieval-augmented generation system: documents chunked into 1000-char windows, embedded via all-MiniLM-L6-v2, indexed in Qdrant, and retrieved through a two-stage top-k retrieval → Cohere reranking pipeline before Gemini generation.
- Enforced grounded responses with inline source citations to eliminate hallucination, and reduced per-query cost by routing to Gemini Flash after token usage profiling — maintaining retrieval precision while cutting latency.

Media Sync Engine | AI Sports Commentary Generator **2025**

- Engineered a multi-model inference pipeline: BLIP extracts scene descriptions from video frames, Llama 3 (via Ollama) transforms them into broadcast-style commentary scripts, and gTTS + MoviePy synchronize narration audio onto highlight reels.
- Designed the FastAPI backend with async video processing endpoints and a Streamlit client supporting drag-and-drop upload, real-time preview, and one-click download of fully narrated sports highlight videos.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C

Backend & APIs: Next.js ,Node.js, Express.js, FastAPI, Django, REST APIs, WebSockets, Authentication

Frontend: React, Vite, Tailwind CSS, Streamlit

AI / ML: TensorFlow, Keras, OpenCV, Scikit-learn, YOLOv8, LangChain, Hugging Face, Deep Learning, Computer Vision, NLP, RAG, LLMs, Generative AI

Databases: PostgreSQL, MongoDB, MySQL, Qdrant

Tools & DevOps: Git, Docker, Postman, Google Colab